

mRNA therapy as primary and bridge therapy for inborn errors of metabolism

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Development of therapies for rare disease involves a cascade of *in vitro* testing, often starting with structural modeling then moving to cell studies, followed by confirmation of *in vivo* efficacy in animal models before first-in-human studies can occur.¹ In the case of inborn errors of metabolism, correction of the underlying enzymatic defect is a critical biomarker for the preclinical studies, but one that is often stymied by lack of an appropriate model. In this setting, naturally occurring large animal models exist for a limited number of enzyme deficiencies, but engineered mouse models are far more commonly used for proof-of-concept studies. However, even mouse models can be challenging, as they sometimes do not reflect the full human phenotype or are so severe that the defect is either lethal *in utero* or deficient animals do not survive long enough post-delivery to be studied. Such has been the case for the X-linked disorder ornithine transcarbamylase (OTC) deficiency. In its most severe form, it presents with overwhelming hyperammonemia with high morbidity and mortality.² Milder patients, especially females, can have later-onset disease with less dramatic episodes of hyperammonemia but still with significant neurologic and intellectual impairment.

Animal models for this condition have been either too mild to accurately reflect severe neonatal disease (the most pressing need for acute therapy), or too severe to serve as an *in vivo* model due to early death.³ Thus, the OTC model described by Zhou et al. in this issue of *Molecular Therapy* represents an important advance in the field.⁴ Their inducible model of OTC deficiency leads to severe deficiency and is amenable for the study of new therapies, in this case lipid nanoparticle encapsulated OTC mRNA.

The model was generated using cre/lox gene editing technology to delete exon 2 of the OTC gene using a tamoxifen inducible promoter to achieve efficient knockout of OTC expression in adult animals. The animals survive and develop hyperammonemia with impaired ureagenesis, chronic symptoms including poor growth, and characteristic pathologic changes on a normal protein diet. Treatment with liver-directed OTC mRNA delivered via nanoparticles was remarkably effective both acutely and with long-term treatment in controlling hyperammonemia and its symptoms. It also reduced accumulation of orotic acid and glutamine, standard clinical biomarkers of the disease. Given the relative effectiveness of the therapy, the apparent safety of other lipid nanoparticle therapies in early clinical trials for inborn errors of metabolism, and a successful phase 1 study of another OTC mRNA therapy in adult patients with OTC deficiency (clinicaltrials.gov study NCT06488313), it seems that this therapy is ready to transition to first-in-human clinical trials.

Development of a potential mRNA therapy for OTC deficiency raises the question of its relative role in treatment vs. other therapies for this condition (Figure 1). Phenylacetate and/or its prodrug phenylbutyrate have been used for over 40 years as an alternative ammonia conjugating agent in the treatment of OTC deficiency, but as babies reach 1 year of age or 10 kg in weight, the drug cannot keep up with the body's need for ammonia removal and mortality rises.⁵ Thus, most severely affected infants receive a liver transplant in the first year of life, trading OTC deficiency for a different set of medical problems related to chronic immunosuppres-

sion.⁶ Liver-directed gene therapy for OTC is currently in clinical trials, but has only been used in adolescents or adults (clinicaltrials.gov study NCT05345171). A second study focused on neonatal OTC is now enrolling (clinicaltrials.gov study NCT06255782). However, a number of concerns about the use of gene therapy in young infants and children have limited its development for this population, including long-term safety, durability of effect, and off-target effects. Here, a role for mRNA as a transition therapy for OTC seems logical, with use for a year or 2 to bridge severe patients to liver transplant or gene therapy. Of course, long-term therapy with mRNA is potentially feasible but depends on the interval needed between infusions to maintain clinical efficacy. It is important to note that while most mRNA therapies currently in clinical trials are liver targeted, effects beyond liver symptoms have been reported. Mice with deficiencies propionyl-CoA carboxylase (the cause of propionic acidemia) and methylmalonyl-CoA mutase (leading to primary methylmalonic acidemia) have shown significant systemic improvements with liver-based mRNA therapies, as have patients with propionic acidemia in early-phase clinical trials.^{7,8} Similarly, treatment of very long and medium-chain acyl-CoA dehydrogenase- (VLCAD and MCAD, respectively) deficient mice with lipid nanoparticle encapsulated mRNA provides recovery of cold tolerance typical of wild-type animals, a phenotype associated with brown fat in mice and muscle in humans.^{9,10} Thus, it is important to remember that liver-based mRNA therapy can have broader physiologic effects, potentially expanding the scope of diseases it can treat.

As mRNA therapies in preclinical development expand, it is appropriate to consider their path to approval by the Food and

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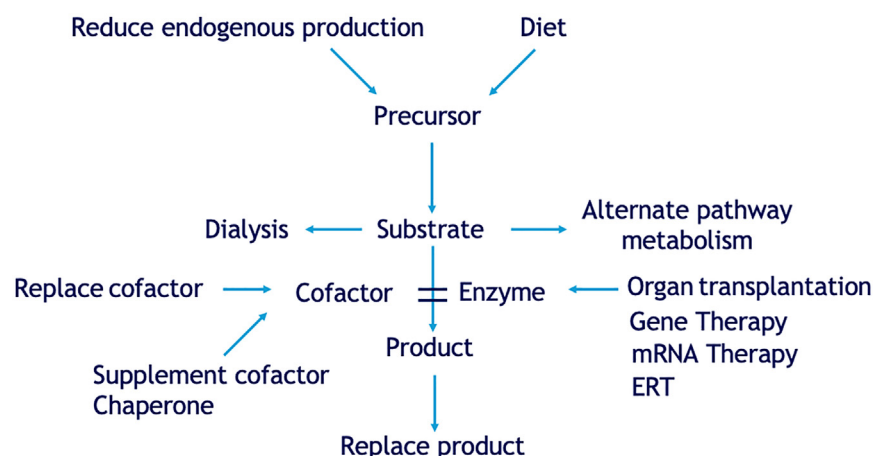


Figure 1. A stylized paradigm for treatment of inborn errors of metabolism

A deficient enzyme leads to an increase in substrate and a decrease in product, either or both of which can lead to cellular metabolic imbalance. Therapies can reduce the flow of substrate to the enzyme, replace the product, or replace the enzyme by organ transplantation, nucleic acid therapy, or enzyme replacement therapy. Mutant enzymes can sometimes be stabilized by small molecule chaperonins or supplementation of excess cofactor(s). ERT = enzyme replacement therapy. Reproduced with permission from the Society for Inborn Errors of Metabolism North American Metabolic Academy curriculum.

Drug Administration (FDA) as medications for human use. Overall, the FDA's current legislative mandates and regulations for drug development in rare diseases are focused on incentivizing research, expediting approval processes, and ensuring that patient perspectives are integral to the development of new therapies.¹¹ These efforts reflect a growing recognition of the unique challenges faced by individuals with rare diseases and a commitment to improving access to treatment. The FDA Rare Diseases Program is designed to streamline the approval process for drugs targeting rare conditions. This program encourages the submission of investigational new drug applications (INDs) and provides guidance on expedited review processes, including Fast Track designation, Breakthrough Therapy designation, and Accelerated Approval pathways. These initiatives aim to reduce the time it takes for patients to access new therapies. In the traditional regulatory environment, each mRNA preparation would be considered a new drug and require a full development path

for approval including all toxicity and safety testing. However, there is significant merit to considering such drugs as a platform where the safety is accepted as proven for any mRNA package so long as the capsid for deliver is identical. While each mRNA drug still would need to prove efficacy for the disease in question, the time and expense to final FDA approval would be reduced substantially. Such a change would require explicit action by the FDA and represent a new paradigm in drug development. Based on its rare disease model, it is likely that the agency indeed has the authority to reach a decision to enable this approach, but it has not yet demonstrated the willingness to do so. If this hurdle is overcome, development of mRNA therapies will be poised to bring these drugs to market for a broad range of disorders including additional inborn errors of metabolism.

DECLARATION OF INTERESTS

J.V. is an investigator in clinical trials sponsored by Moderna Pharmaceuticals.

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